Haltech Wiring Report - Page 1

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

Assigned Resources:

Connections			Assigned Functions
Analogue Voltage Inputs on Nexus R3 VCU:			
	[W], C10	AVI1	Brake Pedal Switch 1 Input, for Brake Pedal Switch
	[W/Y], C11	AVI2	Clutch Switch Input, for Clutch Switch
	[W/GY], C12	AVI3	Coolant Temperature Input, for Coolant Temperature Sensor
	[W/V], C13	AVI4	Oil Pressure Input, for Oil Pressure Sensor
	[W/G], C14	AVI5	Throttle 1 Sensor A, for Drive By Wire
	[W/O], C15	AVI6	Throttle 1 Sensor B, for Drive By Wire
	[W/B], C16	AVI7	Intake Air Temperature Input, for Intake Air Temperature Sensor
	[W/BR], C17	AVI8	Pedal Sensor A, for Drive By Wire
	[W/R], C18	AVI9	Pedal Sensor B, for Drive By Wire
	[LG], C27	AVI10	Oil Temperature Input, for Oil Temperature Sensor
	[LG/B], C28	AVI11	Fuel Pressure Input, for Fuel Pressure Sensor
Digital Pulsed Outputs on Nexus R3 VCU:			
	[V/B], A9	DPO1	Output 1, for Speedometer (Generic Output 4)
	[V/BR], A12	DPO2	Tacho Output, for Tachometer
	[V/R], A14	DPO3	Oil Pressure Light, for Oil Pressure Light
	[V/O], A15	DPO4	Output 1, for Boost Gauge (Generic Output 9)
	[V/Y], A16	DPO5	Check Engine Light Output, for Check Engine Light
	[V/G], A17	DPO6	Boost Control Solenoid, for Boost Control
Sync Pulsed Inputs on Nexus R3 VCU:			
	[GY/BR], C5	SPI1	Vehicle Speed Input, for Vehicle Speed Drive Train Sensor
	[GY/R], C6	SPI2	Vehicle Speed Rear Left Input, for Vehicle Speed Rear Left Sensor

Haltech Wiring Report - Page 2

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

Sync Pulsed Inputs on Nexus R3 VCU:

	[GY/O], C7	SPI3	Brake Pressure Front Input, for Brake Pressure Front Sensor
	[GY/P], C19	SPI5	Override Switch, for Fog Lamps (Generic Output 3)
	[GY/LG], C20	SPI6	Override Switch, for Horn (Generic Output 11)

Knock Inputs on Nexus R3 VCU:

	[B/L], C23	KNOCK1	Knock Sensor 1

Injection Outputs on Nexus R3 VCU:

	[L], A1	INJ1	Injector Output for Cylinder 1
	[L/B], A2	INJ2	Injector Output for Cylinder 2
	[L/BR], A3	INJ3	Injector Output for Cylinder 3
	[L/R], A4	INJ4	Injector Output for Cylinder 4

Ignition Outputs on Nexus R3 VCU:

	[Y/B], A27	IGN1	Ignition Output for Cylinder 1
	[Y/R], A28	IGN2	Ignition Output for Cylinder 2
	[Y/O], A29	IGN3	Ignition Output for Cylinder 3
	[Y/G], A30	IGN4	Ignition Output for Cylinder 4

Half Bridge Outputs on Nexus R3 VCU:

	[BR/B], A19	HBO1	Throttle 1 Motor A, for Drive By Wire
	[BR/R], A20	HBO2	Throttle 1 Motor B, for Drive By Wire
	[BR/G], A21	HBO3	Output 1, for Oil Pressure Gauge (Generic Output 8)
	[BR/P], A22	HBO4	Output 1, for Coolant Temp Gauge (Generic Output 7)
	[P/R], A18	HBO5	CAN Power Output 1, for CAN System
	[P/BR], A25	HBO6	Output 2, for Always On Circuits (Generic Output 1)

Haltech Wiring Report - Page 3

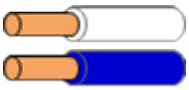
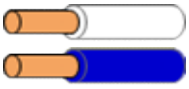
Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

25A High Current Outputs on Nexus R3 VCU:

	[R/L], E1	25A-HCO1	Injector Power Output 1, for Injection System
	[R/Y], E2	25A-HCO2	Ignition Coil Power Output 1, for Ignition System
	[R/O], E3	25A-HCO3	Starter Motor Relay Output, for Engine Starter Control Function
	[R/G], E4	25A-HCO4	Thermofan 1 Output, for Thermofan 1

CAN Ports on Nexus R3 VCU:

	[W], A23, CAN 1 H [L], A24, CAN 1 L	CAN1	Haltech CAN Bus Port 1, for CAN System
	[W], C21, CAN 2 H [L], C22, CAN 2 L	CAN2	Haltech CAN Bus Port 2, for CAN System

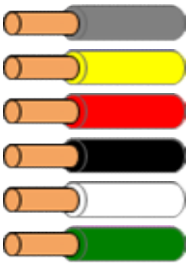
Trigger on Nexus R3 VCU:

	[GY], C1	Trig	Trigger
	[GY/B], C3	Home	Home

Pressure Inputs on Nexus R3 VCU:

		OBPS 1	Manifold Pressure Input, for Manifold Pressure Sensor

Wideband Inputs on Nexus R3 VCU:

	[GY], C29, WBI 1 HEATER+ [Y], C30, WBI 1 INPUT [R], C31, WBI 1 PUMP [B], C32, WBI 1 NERNST [W], C33, WBI 1 HEATER- [G], C34, WBI 1 CAL	Internal Wideband	Wideband O2 1 Input, for Wideband O2 1 Sensor

System References on Nexus R3 VCU:

	[O], C25	+5V Sensor Supply	5V Sensors
	[B/W], C26	Sensor GND	Sensor ground

Haltech Wiring Report - Page 4

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

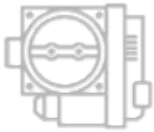
System References on Nexus R3 VCU:

	[O/W], C9	+8V Sensor Supply	8V Sensors
		+12V ECU Power	Battery (+ve)
		Battery Ground	Battery (-ve)
	[B], A10, POWER GROUND [B], A11, POWER GROUND	Power GND Out	Power Ground Out

Do not directly connect any other devices to the ignition switch

	[P], A13	Ign SW Input	Ignition Switch
	[R/L], A26	Ign SW Power	Ignition Switch Power

Other Connections on Drive By Wire:

		Idle	Main Output, for Idle Control

Buttons on Main Keypad (Keypad 2x4 A) :

		Keypad-2x4 A Button 1	Enable / Disable, for Cruise Control
		Keypad-2x4 A Button 2	Set / Coast / -, for Cruise Control
		Keypad-2x4 A Button 3	Resume / Accel / +, for Cruise Control
		Keypad-2x4 A Button 4	Hazard Lights Input, for Indicator Control
		Keypad-2x4 A Button 5	Datalog Switch Input, for Datalogging Switch

Haltech Wiring Report - Page 5

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

Buttons on Main Keypad (Keypad 2x4 A) :

	Keypad-2x4 A Button 6	Thermofan 1 Override Input, for Thermofan 1
	Keypad-2x4 A Button 7	Input, for Boost Trim (Rotary Trim Module 1)
	Keypad-2x4 A Button 8	Engine Bump Input 1, for Engine Bump

Channels:

	Kill Switch Input 1 (GenCond5 Out)	Kill Switch Input 1, for Kill Switch
	Cancel (GenCond6 Out)	Cancel, for Cruise Control

Analogue Voltage Inputs on Boot Power Distro (PD16 Box A) :

	[W], A20	PD16A AVI1	Park Light Input, for Headlight Control
	[W/Y], A21	PD16A AVI2	Head Light Input, for Headlight Control
	[W/GY], A27	PD16A AVI3	Left Indicator Input, for Indicator Control
	[W/V], A28	PD16A AVI4	Right Indicator Input, for Indicator Control

Sync Pulsed Inputs on Boot Power Distro (PD16 Box A) :

	[GY/BR], A14	PD16A SPI1	Override Switch, for Heater (Generic Output 5)
	[GY/R], A15	PD16A SPI2	Start Button Input, for Engine Starter Control Function
	[GY/O], A16	PD16A SPI3	Override Switch, for Wipers (Generic Output 2)
	[GY/Y], A19	PD16A SPI4	Fuel Level 1 Input, for Fuel Level Sensor 1

Half Bridge Outputs on Boot Power Distro (PD16 Box A) :

	[BR/Y], A1	PD16A HBO1	Battery Light Output, for Battery Voltage Light
	[BR/L], A2	PD16A HBO2	Output 1, for Fuel Gauge (Generic Output 10)

Haltech Wiring Report - Page 6

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

8A High Current Outputs on Boot Power Distro (PD16 Box A) :

	[P/R], A3	PD16A 8A-HCO1	Output 1, for Always On Circuits (Generic Output 1)
	[P/BR], A4	PD16A 8A-HCO2	Right Indicator Output, for Indicator Control
	[P/B], A5	PD16A 8A-HCO3	Brake Light Output, for Brake light Control
	[P/O], A6	PD16A 8A-HCO4	Park Lights Output, for Headlight Control
	[P/Y], A7	PD16A 8A-HCO5	Alternator Control Output, for Alternator Control
	[P/G], A8	PD16A 8A-HCO6	Left Indicator Output, for Indicator Control
	[P/V], A9	PD16A 8A-HCO7	Head Lights Output, for Headlight Control
	[P/L], A10	PD16A 8A-HCO8	Output 1, for Heater (Generic Output 5)
	[P/GY], A17	PD16A 8A-HCO9	Output 1, for Horn (Generic Output 11)
	[P/W], A18	PD16A 8A-HCO10	Output 1, for Wipers (Generic Output 2)

25A High Current Outputs on Boot Power Distro (PD16 Box A) :

	[R/Y], E2	PD16A 25A-HCO2	Output 1, for Fog Lamps (Generic Output 3)
	[R/O], E3	PD16A 25A-HCO3	Fuel Pump Relay Output, for Primary Fuel Pump

System References on Boot Power Distro (PD16 Box A) :

		PD16A +12V Power	Battery (+ve)
	[O], A12	PD16A Sensor 5V	Sensor 5V
	[B/W], A11	PD16A Sensor GND	Sensor ground
	[B], A31, POWER GROUND	PD16A Power GND In	Power ground
	[B], A32, POWER GROUND		
	[B], A33, POWER GROUND		
	[B], A34, POWER GROUND		

Do not directly connect any other devices to the ignition switch

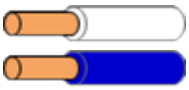
	[P], A13	PD16A Ign SW Input	Ignition Switch Input
	[R/L], A26	PD16A Ign SW Power	Ignition Switch Power

Haltech Wiring Report - Page 7

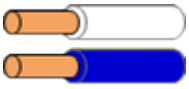
Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

System References on Boot Power Distro (PD16 Box A) :

	[W], A23, CAN H [L], A24, CAN L	PD16A CAN bus	Haltech CAN
	[P], A25	PD16A CAN Power	CAN Power
	[B], A22	PD16A CAN Ground	CAN Ground

System References on Haltech Dash:

	[R], A3	Dash +12V ECU Power	Battery (+ve)
	[B], A4	Dash Battery Ground	Battery (-ve)
	[O], A5	Dash +5V Sensor Supply	5V Sensors
	[B/W], A30	Dash Sensor GND	Sensor ground
	[W], A1, CAN H [L], A2, CAN L	Dash CAN - Main Connector	Haltech CAN

Haltech Wiring Report - Page 8

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

Unassigned Resources:

Connections

Sync Pulsed Inputs on Nexus R3 VCU:

	[GY/Y], C8	SPI4

Knock Inputs on Nexus R3 VCU:

	[B/G], C24	KNOCK2

Injection Outputs on Nexus R3 VCU:

	[L/O], A5	INJ5
	[L/Y], A6	INJ6
	[L/G], A7	INJ7
	[L/V], A8	INJ8

Ignition Outputs on Nexus R3 VCU:

	[Y/BR], A31	IGN5
	[Y/L], A32	IGN6
	[Y/V], A33	IGN7
	[Y/GY], A34	IGN8

Trigger on Nexus R3 VCU:

	[GY], C2	Trig(-)
	[GY/B], C4	Home(-)

Virtual Outputs on Nexus R3 VCU:

Virtual Output No Physical Connection	Virtual Output 1
Virtual Output No Physical Connection	Virtual Output 2

Haltech Wiring Report - Page 9

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

Virtual Outputs on Nexus R3 VCU:

Virtual Output: No Physical Connection	Virtual Output 3
Virtual Output: No Physical Connection	Virtual Output 4
Virtual Output: No Physical Connection	Virtual Output 5
Virtual Output: No Physical Connection	Virtual Output 6
Virtual Output: No Physical Connection	Virtual Output 7
Virtual Output: No Physical Connection	Virtual Output 8
Virtual Output: No Physical Connection	Virtual Output 9
Virtual Output: No Physical Connection	Virtual Output 10

Inertial Measurement Unit (IMU) on Nexus R3 VCU:

	Internal IMU

Channels:

Channel	Override Switch (GenCond1 Out)
Channel	Cancel (GenCond2 Out)
Channel	Disable (GenCond3 Out)
Channel	Set / Coast / - (GenCond4 Out)

Haltech Wiring Report - Page 10

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

Channels:

Channel	Cruise Control Up Alternate Function
Channel	Cruise Control Down Alternate Function

25A High Current Outputs on Boot Power Distro (PD16 Box A) :

	[R/L], E1	PD16A 25A-HCO1
	[R/G], E4	PD16A 25A-HCO4

Analogue Voltage Inputs on Haltech Dash:

	[W], A18	Dash AVI1
	[W/Y], A19	Dash AVI2
	[W/GY], A20	Dash AVI3
	[W/V], A21	Dash AVI4
	[W/G], A22	Dash AVI5
	[W/O], A23	Dash AVI6
	[W/B], A24	Dash AVI7
	[W/BR], A25	Dash AVI8
	[W/R], A17	Dash AVI9

Digital Pulsed Inputs on Haltech Dash:

	[GY/BR], A33	Dash DPI 1
	[GY/R], A32	Dash DPI 2
	[GY/O], A31	Dash DPI 3

Digital Pulsed Outputs on Haltech Dash:

	[V/B], A26	Dash DPO 1
	[V/BR], A27	Dash DPO 2
	[V/R], A28	Dash DPO 3

Haltech Wiring Report - Page 11

Vehicle Name: **Bad Wolf Turbo Mk2**

Product: **Haltech Nexus R3 VCU**

Digital Pulsed Outputs on Haltech Dash:

	[V/O], A29	Dash DPO 4

Buttons on Haltech Dash:

	Button 1
	Button 2
	Button 3
	Button 4
	Button 5
	Button 6